THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Cat's-eye reflectors fixed on roads are used to: 1

(a) absorb light
(b) reflect headlight beams back to the driver
(c) refract light
(d) scatter all light

2. Assertion (A): In a plane mirror, a person who raises the right hand sees the image raise the left hand. Reason (R): A plane mirror produces lateral inversion. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A convex lens of focal length 20 cm is used as a magnifier. The object must be at a distance: 1

(a) infinity **(b)** less than 20 cm
(c) greater than 40 cm **(d)** equal to 40 cm

4. A diverging lens is used in a peep-hole of a door to see: 1

(a) inverted image **(b)** an enlarged image
(c) real image **(d)** a wide view with a smaller erect image

5. Assertion (A): Two parallel plane mirrors facing each other can form an infinite number of images. Reason (R): Light is repeatedly reflected between the mirrors. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The Sun's image is formed by a convex lens at 15 cm from the lens. Its focal length is: 1

(a) infinity **(b)** 30 cm **(c)** 7.5 cm **(d)** 15 cm

7. A student places an object 25 cm from the eye to see it clearly. This distance is called the: 1

(a) far point **(b)** optical centre **(c)** near point **(d)** focal point

8. Assertion (A): A real image can be obtained on a screen. Reason (R): Real images are formed when reflected or refracted rays actually meet. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A pencil placed at 5 cm from a plane mirror moves 3 cm closer. The distance between the pencil and its image becomes: 1

(a) 8 cm **(b)** 10 cm **(c)** 2 cm **(d)** 4 cm

10. For a person who needs concave spectacles, the eyeball is likely: 1

(a) of correct length **(b)** missing lens

(c) too short **(d)** too long

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an object appear in colour when seen through a convex lens but not a prism? 2

12. Why is the image in a concave mirror at 2f the same size as the object? 2

13. The focal length of a mirror is 7.5 cm. Find the radius of curvature and the distance at which the image of the Sun forms. 2

14. State the effect on the number of images when the angle between two mirrors is reduced. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the image formed by a concave mirror is smaller when the object is far away and larger when nearer F. 3

16. A student wants a mirror to see the back of her head. How can she use two plane mirrors? 3

17. Explain why the letters on the front of an ambulance read 'ECNALUBMA'. 3

OR

State why a convex lens used as a magnifier gives an enlarged image only for objects close to it. 3

18. Explain why we get multiple images in thick glass mirrors. 3

OR

Explain how the shape of the eye lens changes to focus on near and distant objects. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain, with reasons, why a dentist's mirror, a car headlight and a convex shop mirror have different shapes. Compare the nature of images formed. 5

OR

Explain how the image of a distant tree is formed on the retina of the eye, naming the parts involved, the nature of the image and how the brain interprets it. 5

20. A student sets up a candle, a convex lens and a screen. Describe how she can obtain a real image bigger than, same as and smaller than the object, using positions relative to f and 2f. Include measurements to verify. 5

OR

Discuss any five innovations or applications of reflection and refraction in technology and health care. 5

END OF QUESTION PAPER — Total: 40 Marks